Baseline Is Not Health

Epistemic baselines, intervention epochs, and the danger of teaching condition-monitoring systems that the first stable state is a healthy state

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Thesis. A newly observed stable machine state is evidence of repeatability, not evidence of health. Condition-monitoring systems should preserve that distinction explicitly, and maintenance interventions should create new temporal epochs rather than silently rewriting the baseline.

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Abstract

Condition-monitoring systems commonly compare current measurements with baselines, thresholds, learned distributions, or reference windows. This is useful but hides a dangerous epistemic shortcut: a stable initial distribution is often treated operationally as if it represented healthy equipment. For a newly installed monitor on an asset with unknown history, that conclusion is not warranted. The asset may already contain bearing damage, imbalance, looseness, thermal stress, misalignment, or an abnormal operating regime that is stable enough to be learned. This paper proposes an explicit separation between **observed baseline** and **health-validated baseline**, together with intervention-bounded temporal epochs that preserve pre- and post-maintenance behavior without retroactively declaring either healthy. The approach complements established condition-monitoring architectures such as ISO 13374 and ISO 17359, and addresses a related but distinct problem from statistical concept drift: the meaning of a distribution can change because the machine, its operating regime, or our knowledge of its condition changed. Maintenance events are therefore treated as evidence-bearing interventions, not merely annotations. The paper develops a state model, data architecture, evaluation criteria, counterarguments, and implementation path for edge condition-intelligence systems. SynapticSteel is used only as a concrete architectural context; the proposed epoch semantics are not represented as a completed production capability.

Evidence boundary. Standards and cited literature establish the need for structured condition-monitoring pipelines, traceable timing, drift handling, and maintenance-aware analysis. The specific epistemic-baseline and intervention-epoch architecture developed here is a proposal and inference from those requirements, not a claim that the cited standards prescribe it.

1. The hidden assumption inside a baseline

A baseline answers a statistical question: *what does this asset look like under some observed set of conditions?* Health is a different claim: *is the asset in an acceptable physical condition?* The two questions overlap only when independent evidence connects the observed state to health. A monitor installed on Tuesday cannot infer from Wednesday's repeatability that the machine was healthy on Tuesday. It can infer only that the measured behavior was sufficiently stable to characterize.

This distinction matters because many practical detection methods use reference windows, learned normal distributions, alarm bands, or distance from a baseline. ISO 13374 provides an open reference architecture for condition-monitoring and diagnostics data processing, while ISO 17359 provides general guidance for establishing condition-monitoring programs [1,2]. Neither fact eliminates the commissioning problem: the semantics attached to a learned reference still depend on what is known about the asset when that reference is captured.

Recent machine-condition research also treats concept drift as a real operational problem. Sojo, Bielza, and Larrañaga propose an unsupervised Bayesian-network drift detector that compares a fixed reference window with a sliding window and adapts the model after detected drift [3]. That is valuable, but drift detection and health validation are not the same operation. A detector can correctly discover that the distribution changed while remaining unable to say whether the old state, new state, both states, or neither state is healthy.

Object	Defensible claim	Claim it must not imply
Initial observed baseline	Behavior was observed and sufficiently stable under recorded conditions.	The machine was healthy.
Health-validated baseline	Observed behavior is linked to independent evidence supporting acceptable condition.	The state will remain healthy indefinitely.
Drift event	The monitored distribution changed beyond the selected detector's criterion.	The change is a fault.

Object	Defensible claim	Claim it must not imply
Maintenance event	A human intervention occurred at an approximate/known time with recorded scope.	The intervention caused every subsequent change.
Post-maintenance epoch	Behavior after intervention can be compared with the preceding epoch.	The post-maintenance state is automatically healthy.

Table 1. The architecture should preserve the difference between observation, validation, drift, intervention, and causal attribution.

2. Why maintenance creates an epistemic boundary

Maintenance is not merely another timestamp in the telemetry stream. Replacing a bearing, tightening a coupling, aligning a shaft, changing lubrication, cleaning a heat exchanger, modifying a process setpoint, or even opening and reassembling a machine can alter the data-generating process. In statistical language, the post-maintenance distribution may differ. In physical language, the machine itself or its operating context may have changed. In epistemic language, the monitor has new evidence that should change how earlier and later observations are compared.

Predictive-maintenance research increasingly combines telemetry with maintenance and failure records. A 2026 Scientific Reports study, for example, describes a benchmark organized across telemetry, errors, maintenance, failures, and machine metadata, explicitly aligning maintenance interventions with time-series records [4]. Work on prescriptive maintenance likewise argues that sensor data, repair records, maintenance histories, operational parameters, and domain expertise are needed to build useful maintenance knowledge [5]. These approaches support a broader conclusion: intervention records are part of the evidence model, not administrative decoration.

The proposed architecture therefore introduces an **epoch boundary**. An epoch is a bounded period during which observations are interpreted under a declared baseline context. A maintenance event can close the previous epoch and open a new one, but it does not erase the old epoch. This preserves a powerful future comparison: if a machine's original stable signature later reappears and is associated with known degradation, the system can reinterpret the historical baseline as 'stable but degraded' without rewriting the original evidence.

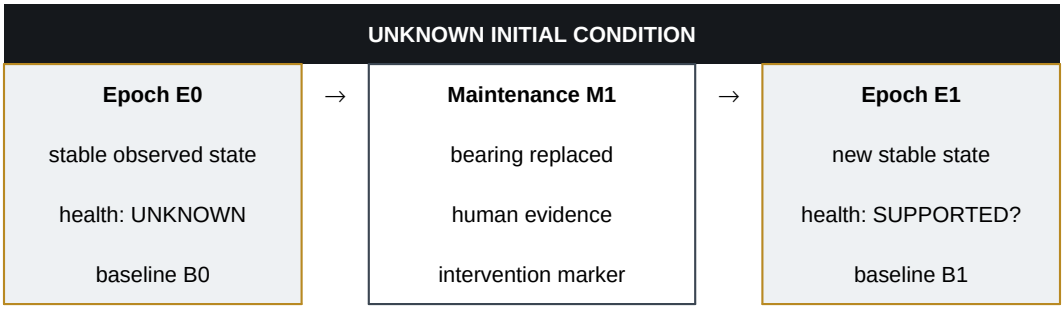


Figure 1. Maintenance creates a comparison boundary. Neither E0 nor E1 is declared healthy solely because it is stable.

3. Proposed architecture: baseline semantics as first-class state

The minimal implementation requires more than storing a rolling mean. It needs explicit state objects whose semantics survive model replacement and software upgrades. The proposed record set is intentionally model-agnostic.

Record	Minimum content	Purpose
Observation	source, asset, channel, value/features, acquisition time, timing uncertainty, quality	Preserve what was measured.
Baseline	baseline_id, epoch_id, feature definition, operating context, window, model/version	Preserve the reference actually used.

Record	Minimum content	Purpose
Health assertion	asset/epoch, status, evidence source, issuer, confidence, valid interval	Keep health claims separate from statistical stability.
Intervention	asset, approximate/exact time, work performed, parts/actions, human observations	Mark a possible physical state transition.
Epoch	start/end, predecessor, trigger, active baseline set, operating regime	Bound comparisons without deleting history.
Comparison	from_epoch, to_epoch, features, effect sizes, uncertainty, interpretation	Make pre/post differences inspectable.

Table 2. Proposed records for separating physical evidence from interpretation.

A compact state rule is:

stability(E) does not imply health(E)

health(E) requires an evidence-bearing assertion whose provenance is independent of the mere existence of a stable reference distribution.

4. Time is part of the evidence

Pre/post intervention comparison fails if event time is treated as perfectly known when it is not. A technician may enter 'around 14:30'; a BLE sensor may timestamp at receipt rather than acquisition; a camera pipeline may buffer frames; an edge service may restart with a clock offset. The system should therefore store not only a timestamp but, where material, its uncertainty and clock domain.

NIST's work on IEEE 1588 emphasizes that synchronized component clocks allow sensor data to be timestamped at the source and correlated after acquisition [6]. NIST also notes that timing traceability is meaningful only with a documented uncertainty chain [7]. For industrial condition intelligence, this suggests a practical rule: correlation windows should widen or confidence should fall when timestamp uncertainty is large. A maintenance marker entered after the fact should not be treated as a nanosecond-precise causal boundary.

This is especially relevant for multimodal edge systems. Vibration, acoustic, thermal, ToF, network, and maintenance records can operate at radically different cadences. The architecture should preserve acquisition time, ingest time, clock source, and uncertainty rather than collapsing them into a single apparently precise timestamp.

5. Competing approaches and why they are insufficient alone

Fixed thresholds. Standards-based alarm thresholds are valuable where applicable, but they do not solve asset-specific unknown initial condition. A machine can remain below a generic threshold while still differing materially from its own post-repair state.

Rolling or adaptive baselines. Adaptation reduces false alarms under legitimate drift, but aggressive adaptation can normalize degradation. A baseline that continuously chases the machine can erase the very history needed to recognize slow deterioration.

Concept-drift detectors. Drift methods identify distribution change and may adapt models [3]. They do not, by themselves, assign physical meaning or distinguish beneficial maintenance-induced change from degradation.

Digital twins and PHM models. Digital-twin-driven PHM can integrate state estimation, diagnosis, and prognosis [8]. However, a twin still requires trustworthy initialization and update semantics; a sophisticated model does not remove the need to represent what is known versus assumed.

Supervised failure labels. Confirmed failures are powerful ground truth, but they are sparse, delayed, and often component-specific. A condition-intelligence system needs useful semantics before enough failures accumulate to train a strong supervised model.

6. Relevance to SynapticSteel and TelePod

SynapticSteel's TelePod work provides a natural test environment because it combines multiple physical observation channels with local processing and temporal-model experiments. The broader design direction has emphasized observation, condition intelligence, evidence, and human maintenance decisions rather than autonomous repair.

For publication accuracy, the maturity boundary is important. **Implemented/prototyped elements** include physical sensor collection and service-level pipelines on TelePod-class edge hardware, along with work on normalized observations, event transport, and temporal/model layers. **Experimental elements** include multimodal fusion and several temporal/model approaches. **Proposed in this paper** are the explicit unknown-initial-condition health semantics, intervention-bounded epochs, health-assertion records, and automated pre/post epoch comparison as a unified architecture. They should not be described as a completed production feature until implemented and verified.

A useful deployment sequence would begin with semantics, not AI: create epoch IDs; record maintenance markers; label the commissioning epoch as `health=unknown`; retain baseline versions; and expose pre/post comparisons. Only after that substrate is reliable should models be allowed to infer recurring signatures or propose that an earlier stable epoch was likely degraded.

7. Evaluation methodology

The proposal is testable. A prototype should be evaluated against intervention-rich datasets and controlled bench experiments rather than judged by dashboard appearance. The primary question is whether preserving epistemic state improves diagnosis without increasing unsupported health claims.

Question	Candidate metric
Does commissioning avoid false health claims?	fraction of new assets correctly retained as health=unknown until external evidence arrives
Can intervention effects be reconstructed?	percentage of maintenance events with valid pre/post epoch comparisons
Does adaptation erase degradation?	detection delay and false-normalization rate under gradual drift
Can old baselines be reinterpreted?	successful supersession/reclassification without loss of original evidence
Does timing uncertainty alter confidence?	calibration of correlation confidence versus known timestamp error
Does maintenance context improve diagnosis?	change in time-to-correct-hypothesis and false diagnosis rate with/without intervention records

8. Limitations and counterarguments

Epoch proliferation. Every intervention or operating change cannot create an unmanageable forest of baselines. Epoch creation needs policy: some events should be annotations, some should create sub-epochs, and some should force a new reference model.

Human records are imperfect. Maintenance timestamps and descriptions may be approximate, incomplete, or wrong. Human records should be preserved as evidence with provenance, not promoted to infallible ground truth.

Operating regime can dominate condition. Load, speed, product, ambient temperature, and process state can change sensor distributions more than physical degradation. Epoch semantics therefore need operating-context

dimensions or regime-conditioned baselines.

More memory can create more confusion. Retaining every baseline indefinitely is not automatically useful. Systems need retention, summarization, and supersession policies while preserving enough raw evidence for important claims.

Causal language must remain conservative. A post-maintenance improvement is evidence consistent with an intervention effect, not proof of causation unless the experimental or operational design supports that conclusion. Contemporary prescriptive-maintenance research is explicitly exploring causal discovery, but observational industrial data remains vulnerable to confounding [5].

9. Conclusion

A baseline is a memory of observation. Health is a claim about physical condition. Treating those as synonyms is a small semantic shortcut with large consequences: a monitoring system can learn a damaged machine as 'normal,' adapt toward slow degradation, or overwrite the evidence that a repair materially changed behavior.

The proposed remedy is deliberately architectural rather than model-specific. Preserve unknown initial condition. Separate statistical stability from health assertions. Treat maintenance as an evidence-bearing intervention. Bound comparisons with temporal epochs. Preserve old baselines instead of silently replacing them. Carry timing uncertainty into correlation. Let models propose interpretations, but require provenance for stronger claims.

For edge condition intelligence, this may be a more important foundation than adding another anomaly detector. A system that knows the limits of what its baseline means is harder to fool by a stable fault.

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Related SynapticSteel Technical Papers

STP-001 - Failure Metabolism. Why persistent machine intelligence should digest failure into structured memory instead of merely logging it. Discussion draft, August 21, 2026. STP-002 extends the evidence-memory argument into physical condition monitoring: the system must preserve not only failures, but the epistemic status of the baselines

against which future evidence is interpreted.

Publication note

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